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Preface

Preface.

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Contents

Abstract	3
Abstract (in Finnish)	4
Preface	5
Contents	6
Abbreviations	7
Acknowledgements	8
1 Introduction	9
2 Background	10
2.1 Circular economy and digitalization	10
2.2 Circularity in the Finnish construction industry	11
2.3 Motiva’s data platform and ecosystem for material circulations	13
2.4 The potential of pre-demolition audits	13
3 Model and methods	15
3.1 Data	15
3.2 Features	15
3.3 Prediction targets	15
3.4 Modeling approach and algorithms	15
4 Results	16
4.1 Methods and results of model testing	16
5 Discussion	17
5.1 Meaning of modeling results	17
5.2 Roadmap for data aggregation and preservation	17
5.3 Future modeling possibilities	17
6 Summary	18

Abbreviations

API	application programming interface
CE	circular economy
GBC	Green Building Council
GHG	greenhouse gas

Acknowledgements

1 Introduction

2 Background

This chapter establishes the reasoning behind the urgent need for a global transition towards a circular economy (CE) and describes the role digitalization plays in the process. It takes the Finnish construction industry as an example of this transition, and specifies the key challenges and opportunities of driving the change with data and digital solutions.

2.1 Circular economy and digitalization

It is no news that the ongoing environmental crises, such as global warming and loss of biodiversity, pose a large part of the most significant threats of our lifetime. The challenge of steering humankind towards a sustainable way of life is monumental, and requires extraordinary feats, as well as cross-disciplinary coordination and co-operation from all entities ranging from individual people to businesses, industries, countries and international alliances.

In 2020, the Finnish Innovation Fund Sitra conducted a study on the megatrends for the beginning decade [2], identifying five different, but intertwined global megatrends, three of which are the following:

- *The urgent need for ecological reconstruction*
- *The redefinition of the economy*
- *Technology is embedded in everything*

Getting the interactions between technology and economy to work in favour of fighting environmental crises is an incredibly complex, yet critical challenge. In other words, it is safe to state that the two latter trends play a key role in the way we drive ecological reconstruction in the years to come.

The culprits behind our environmental crises are manifold, but at the end of the day, much of the problem boils down to how we use the natural resources our planet provides us with. Already in 2019, resource extraction and processing made up approximately half of global greenhouse gas (GHG) emissions, in addition to more than 90% of biodiversity loss and water stress. [9] It has become evident, that the sun has set on the old linear economy model. We can not afford to extract, use and dispose of resources with the pace we have reached due to assumptions of limitless economic growth with little to no regard to external costs. The concept of a circular economy presents an alternative model to this, where product lifetime and reusability is maximized, production waste is minimized, and business models based on sharing and reusing products are implemented. The potential in these strategies is enormous: according to a Material Economics study from 2018 [4], the transition to a more circular economy could reduce GHG emissions from heavy industry in the EU by 56% by 2050. Contrary to a common misconception, this shift does not need to happen at the cost of economic growth as a whole. Aligning technologies and business models to tackle wasteful value creation chains and shift towards a more

circular model would actually create new markets, and boost employment, lifting EU's GDP by 7% and yielding annual benefits of up to €1,8 trillion by 2030. [5]

As Sitra states, technology is embedded in everything, and the transition to CE is, of course, no exception. The breathtaking pace digitalization has developed with during the last decades has presented us with countless possibilities and opportunities, but also with its own set of challenges. For example, with increasing amounts of digital solutions and services increases also the demand for energy. Hence, it would be naïve to assume digitalization to be a "silver bullet" that directly solves the problems related to implementing CE models, as stated in a 2019 study by the European Policy Centre (EPC). [6] That being said, the study focuses on establishing the role of digitalization as a key enabler of CE - as long as it is properly managed. They divide approaches to use digitalization to accelerate the transition to CE into three categories:

1. Improving knowledge, connections and information sharing
2. Making business models, products and processes more circular
3. Strengthening the roles of citizens and consumers

Within the scope of this thesis, we will mainly focus on category 1, exploring the ways proper data aggregation, utilization and sharing can support businesses in adapting more circular models. Specifically, we will look at the construction industry, as it is responsible for almost half of global raw material extraction and a quarter of global GHG emissions. [3]

2.2 Circularity in the Finnish construction industry

When reviewing the environmental impact of the construction industry, it is essential to divide carbon emissions into three categories according to the phases in the lifecycle of a building:

1. **Upfront carbon** (materials production and construction phase)
2. **Operational carbon** (use phase, energy consumption)
3. **End of life carbon** (demolition phase, demolition waste handling)

These are defined by the World Green Building Council (WorldGBC) in their 2019 report "Bringing embodied carbon upfront" [1]. The same study finds that with the recent advances in the field of renewable energy, operational carbon emissions are reduced significantly. Consequently, upfront and end of life carbon will grow quickly in proportion. In particular, the WorldGBC predicts that upfront carbon will be responsible for up to half of the whole carbon footprint of new construction between the the present and 2050. A similar trend can be seen in Finland, as well. Up until 2020, approximately 80% of the Finnish built environment's total lifecycle emissions came from operational carbon. [7] With low-carbon energy gaining foothold in the Finnish energy market, focus is shifting increasingly towards reducing upfront and

end of life carbon by transitioning to a more circular operating model. The Finnish government has recently taken the lead in driving this transition across industries in Finland. In the spring of 2021, they established a strategic programme [10] to pave the way towards a new Finnish economy based on CE by 2035. The programme dedicates a significant amount of focus on the construction industry, and specifies several action points to promote CE in its context, including financial incentives, digitalization and data management.

There are, however, plenty of challenges to overcome. Especially crucial is the disconnection between supply and demand in the Finnish recycled construction product market. On a demolition site, the prime contractor owns all of the materials (although the responsibility of demolition waste handling lies with the demolition contractor, if the demolition is outsourced by the prime contractor), and therefore is also responsible for processing all of the material streams released during demolition accordingly, including all related costs and taxes. On the other hand, there are plenty of costs embedded in the process of preparing the materials from extraction to reuse, and said costs often outweigh the costs of disposing of these materials, which makes it less appealing for contractors to salvage them for reuse (some components can be extracted pre-demolition and used again as such) and recycling (products that are not feasible for reuse are processed into new products). It can thus be argued, that there exists at least a potential financial incentive for the contractors to sell as much of the released materials as possible, to be used in new construction. The challenge is to minimize the costs of processing used construction products and demolition waste for reuse and recycling. The problem described above also has the effect of making virgin materials often more attractive price-wise, which hinders the demand for used and recycled construction products, and makes adapting a more circular model difficult from the point of view of procurement as well.

This problematic is clearly a key obstacle, as it effects negatively both the supply and demand sides of the non-virgin construction product market. This has also been recognized by the Finnish government, as it has recently set in motion investigation projects to e.g. identify ambiguities in construction product qualification procedures. [11] As an example result, the Finnish Ministry of the Environment released a policy brief [12] on verifying the feasibility of reused materials in June 2022. It states that it is sufficient to check for feasibility with construction site-specific verification, i.e. a CE-marking is not required for construction products that are reused as such.

However, even if there was a strong financial incentive to favor non-virgin materials, a significant challenge remains aligning demand with supply on a temporal level. Currently, contractors are able to release information of available materials only as soon as they become available, as the demolition project proceeds. Because storage possibilities for these materials are few and often costly, the time window to find a buyer is far too narrow. To counter this challenge, Finnish officials are working on a series of databases and registers that would together provide a centralized data storage system for information on demolition projects and plans, construction product characteristics and material streams, to name a few. These systems are planned to come with diverse APIs to enable private entities to develop their own

services and platforms to utilize all of the data stored in these public systems. The following section will introduce one such project, in which the author is also involved through his employment.

2.3 Motiva's data platform and ecosystem for material circulations

In order to make the raw data from the existing and upcoming registers for construction materials and demolition waste described in the previous section available and usable to companies and consumers, it is vital that different services are developed to refine and process the data. Even though work on the state-wide registers is still in progress, the Finnish state-owned company Motiva Ltd. is already working on a platform for construction material circulation. Motiva is a company of experts specialized in providing organizations and consumers with information, solutions and services on how to use energy and materials more efficiently and sustainably. With this platform, it is Motiva's aim to build a hub that collects information from various sources under a single platform, and acts as a meeting place for supply and demand of non-virgin construction materials.

At the moment, this platform is still in the planning phase. Motiva has developed the concept for it, and it is currently being validated and further developed in two different demolition projects. The first one takes place in Vattuniemi, Helsinki, where 16 buildings are set to be demolished during 2022-23. The other demolition site is located in the Tampere area, where especially the reuse and circulation of ashes as a construction product is being explored. These projects are vital in gathering field experience on the challenges and requirements of material circulation. Specifically, Motiva seeks to pilot planned features of their platform, and to gain knowledge on how exactly predictive, data-driven demolition planning supports a circular economy in the construction industry.

2.4 The potential of pre-demolition audits

A critical part of shifting towards predictive demolition planning are the procedures that guide the mapping of the demolition site. These include mapping of harmful, but also reusable and recyclable materials that the building to be demolished consists of. Historically, only the mapping of asbestos and other harmful materials has been required by law, but nowadays it is increasingly common to do a broader material mapping. The most extensive mapping process is the pre-demolition audit (in Finnish "purkukartoitus"), which also includes, for example, reusable movables and recommendations on how to reuse, recycle or dispose of the material streams released during demolition. [8] These audits are at the moment voluntary, but officials are constantly planning ways to encourage real estate owners and contractors to use them as a tool to increase material circulation. This is slightly challenging, since a more extensive audit is also more expensive to the client. It is, however, crucial from the point of view of CE, that pre-demolition audits are carried out as carefully as

possible, because gaining knowledge of future material streams is necessary in order to align demand with supply in the non-virgin material market.

Carrying out pre-demolition audits, and especially estimating the amounts of different materials released, is hard to do accurately. All experts interviewed for this thesis agreed on this, and stated several possible reasons for the significant errors that occur even in state-of-the-art audits. Some of the material amounts are especially hard to predict, because they might often be hidden from plain sight in the foundations of the building in question. An example of this kind of material is mineral wool, which is often used as an insulator. Especially in the case of old buildings, where the quality of material documentation might be suboptimal, these kind of "hidden" materials can be nigh unto impossible to predict. It might also happen, that some renovations or other changes during the building's lifetime have not been properly documented, which causes some material streams to come as a surprise during demolition. Finally, it is also natural that clients want to minimize the costs of a demolition project, and the pre-demolition auditors may have to compromise extensiveness and accuracy of the audit in order to stay within the budget.

Given the potential of pre-demolition audits from a CE point of view, and the nature of the challenges described above, it is evident that making the audit process more data-driven holds significant possibilities to develop construction material circulation. The obvious bottleneck at the moment, however, is the lack of data. There are no standardized, mandatory procedures to store pre-demolition audit, material mapping, or realized material stream data. This leads to the relatively small amount of feasible data to be scattered all over different stakeholders, and to have little to no quality guarantees. This challenge is at least partly being addressed by the projects described in sections 2.2 and 2.3, but this thesis will contribute to the ways of data aggregation on a more detailed level, and explore the possibilities of utilizing this data in the future.

3 Model and methods

3.1 Data

3.2 Features

3.3 Prediction targets

3.4 Modeling approach and algorithms

4 Results

4.1 Methods and results of model testing

5 Discussion

5.1 Meaning of modeling results

5.2 Roadmap for data aggregation and preservation

5.3 Future modeling possibilities

6 Summary

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